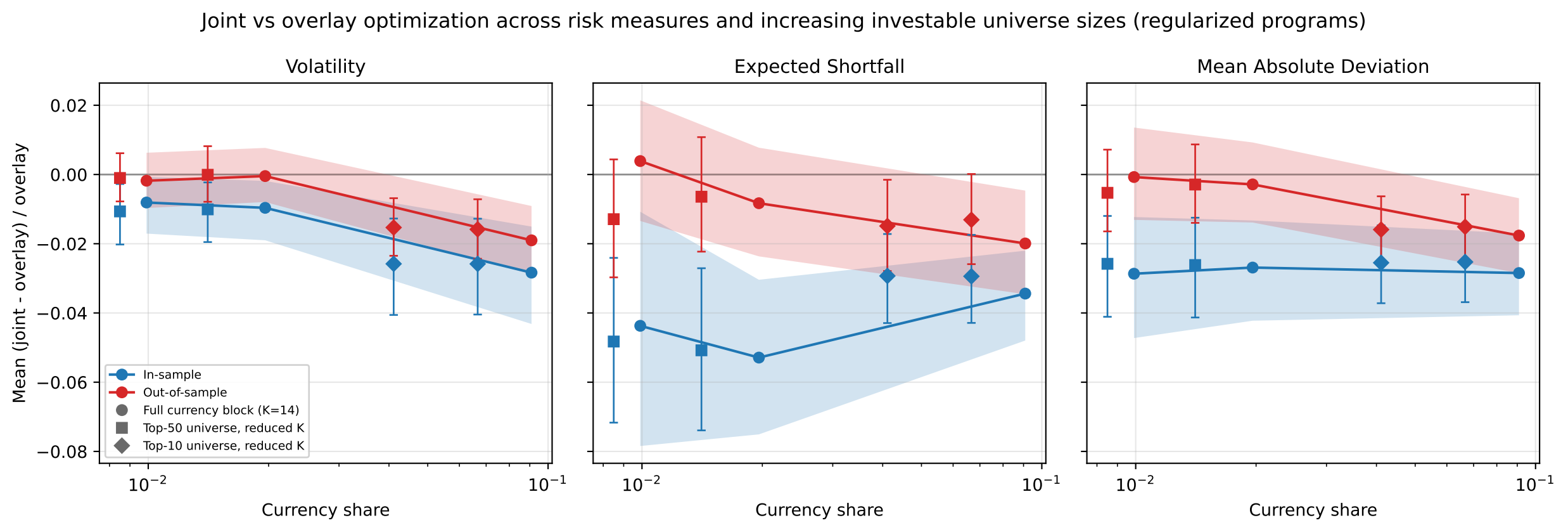
Joint vs overlay optimization across risk measures and increasing investable universe sizes (regularized programs)
Volatility
Expected Shortfall
Mean Absolute Deviation
Mean (joint - overlay) / overlay
0.02
0.00
−0.02
−0.04
−0.06
−0.08
In-sample
Out-of-sample
Full currency block (K=14)
Top-50 universe, reduced K
Top-10 universe, reduced K
10^{-2}
10^{-1}
Currency share